Case Study A — Exploratory Data Analysis

ZeMA electromechanical cylinder dataset (Dorst et al. 2021), doi:10.5281/zenodo.5185953

This notebook explores the **raw input data** used by `case_study_A_analysis.py` — 11 sensor channels × 4,766 run-to-failure cycles, each reading shipped with its own metrologically evaluated standard uncertainty — before any modeling happens. It does **not** reproduce the pipeline (MLP training, LPU/Monte Carlo propagation); see `case_study_A_analysis.py` and `case_study_A_report.md` for that.

Prerequisites

- Run with the project's `zema_env` kernel (*Python (case_study_A - zema_env)*), the same environment used by the pipeline (`pip install -r requirements.txt`), plus `pandas` and `seaborn` for this notebook.
- The dataset HDF5 must already be downloaded/cached. The first `ZeMASamples(...)` call downloads ~2.67 GB from Zenodo if it isn't cached yet — see `README.md` §0–2.

In [1]:
```
import sys
from pathlib import Path

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sys.path.insert(0, str(Path.cwd()))
from case_study_A_analysis import Config, load_zema  # reuse the
pipeline's own loader

sns.set_theme(style="whitegrid")
plt.rcParams["figure.dpi"] = 100
pd.set_option("display.precision", 4)
```

1. Load the dataset

Loads at the same resolution the pipeline uses (`Config` defaults: 4,766 cycles × 100 datapoints/sensor/cycle → 1,100 raw features per cycle). If a previous pipeline run already populated the `pooch` cache, this is near-instant; otherwise it triggers the one-time Zenodo download.

Loading [MathJax]/jax/output/CommonHTML/fonts/TeX/fontdata.js

In [2]:
```
cfg = Config()
X, UX, y = load_zema(cfg)
print(f"X  (values):        {X.shape}")
print(f"UX (uncertainties): {UX.shape}")
print(f"y  (target):         {y.shape}, range [{y.min():.3f},
{y.max():.3f}]")
```

```
X  (values):        (4766, 1100)
UX (uncertainties): (4766, 1100)
y  (target):         (4766,), range [0.000, 1.000]
```

```
/Users/emanuelebuchicchio/repos/case_study_a/zema_env/lib/python3.10/site-
packages/tqdm/auto.py:21: TqdmWarning: IProgress not found. Please update
jupyter and ipywidgets. See
https://ipywidgets.readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
```

2. Channel layout

`X` / `UX` pack 11 sensor channels side by side, each occupying `cfg.datapoints_per_cycle` (100) consecutive columns, in the order the ZeMA HDF5 stores them (`zema_emc_annotated.dataset.ZEMA_QUANTITIES` , with `Acceleration` and `Motor_Current` expanded over their 3 axes/phases).

In [3]:
```
CHANNEL_NAMES = [
    "Acceleration plain bearing", "Acceleration piston rod", "Acceleration
ball bearing",
    "Active current", "Axial force",
    "Motor current phase 1", "Motor current phase 2", "Motor current phase
3",
    "Pneumatic pressure", "Sound pressure", "Velocity",
]
D = cfg.datapoints_per_cycle
assert len(CHANNEL_NAMES) * D == X.shape[1]


def channel_slice(i):
    return np.s_[:, i * D:(i + 1) * D]


for i, name in enumerate(CHANNEL_NAMES):
    print(f"{i:2d}  cols [{i * D:4d}:{(i + 1) * D:4d}]  {name}")
```

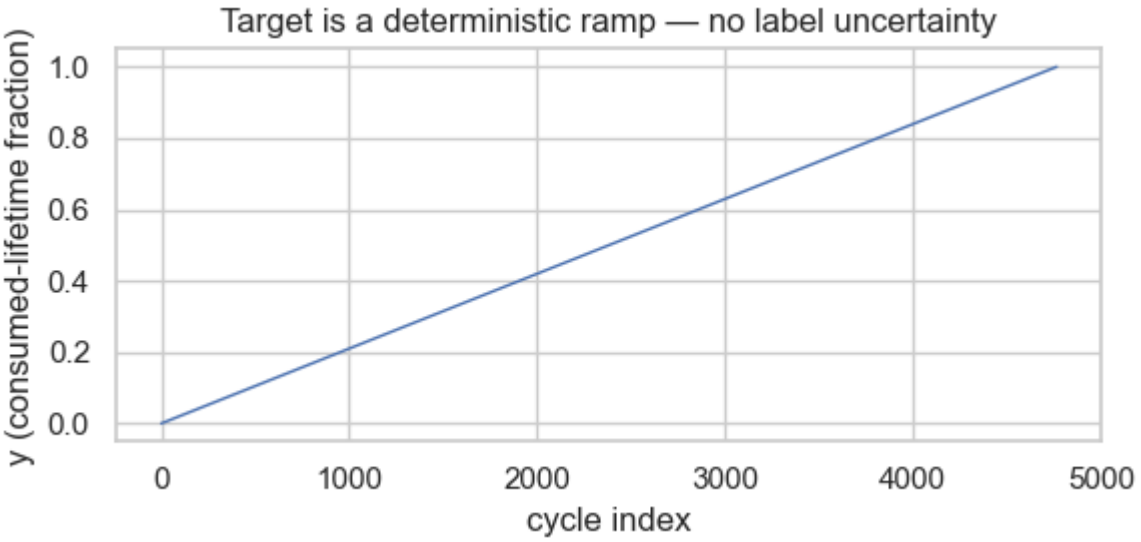
```
 0  cols [   0: 100]  Acceleration plain bearing
 1  cols [ 100: 200]  Acceleration piston rod
 2  cols [ 200: 300]  Acceleration ball bearing
 3  cols [ 300: 400]  Active current
 4  cols [ 400: 500]  Axial force
 5  cols [ 500: 600]  Motor current phase 1
 6  cols [ 600: 700]  Motor current phase 2
 7  cols [ 700: 800]  Motor current phase 3
 8  cols [ 800: 900]  Pneumatic pressure
 9  cols [ 900:1000]  Sound pressure
10  cols [1000:1100]  Velocity
```

Loading [MathJax]/jax/output/CommonHTML/fonts/TeX/fontdata.js

3. Target variable `y`

`y_k = k / N` is the consumed-lifetime fraction, defined purely from the cycle index — not measured — so by construction it is a perfect ramp from 0 to 1 and carries zero label uncertainty. `RUL = 1 − y`. Plotted only to confirm the construction.

In [4]:
```
fig, ax = plt.subplots(figsize=(6, 3))
ax.plot(y, lw=1)
ax.set_xlabel("cycle index")
ax.set_ylabel("y (consumed-lifetime fraction)")
ax.set_title("Target is a deterministic ramp — no label uncertainty")
plt.tight_layout()
plt.show()
```



4. Per-channel value distributions

For each channel, collapse the 100 within-cycle readings to their per-cycle mean and look at the distribution across the 4,766 cycles.

Loading [MathJax]/jax/output/CommonHTML/fonts/TeX/fontdata.js

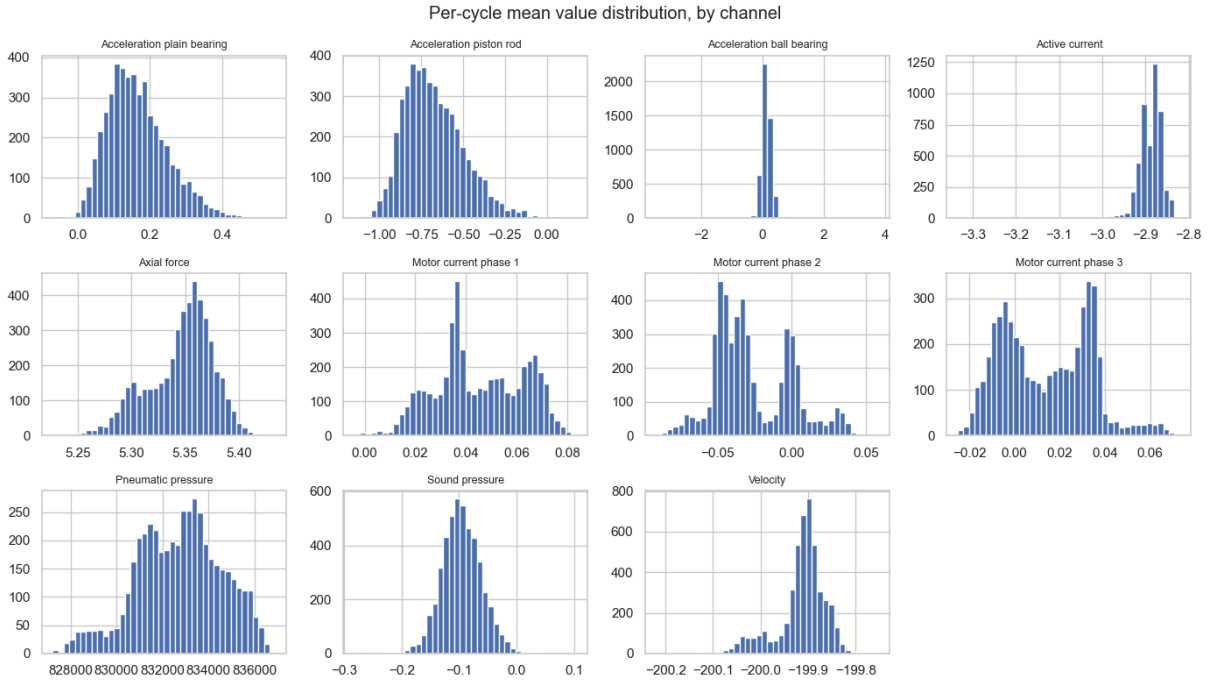
In [5]:
```
df_mean = pd.DataFrame(
    {name: X[channel_slice(i)].mean(axis=1) for i, name in
enumerate(CHANNEL_NAMES)}
)
df_mean.describe().T
```

Out[5]:

	count	mean	std	min	25%	50%
Acceleration plain bearing	4766.0	0.1649	0.0842	-0.0705	0.1039	0.1541
Acceleration piston rod	4766.0	-0.6745	0.1783	-1.1478	-0.8076	-0.6983
Acceleration ball bearing	4766.0	0.1149	0.1750	-3.4820	0.0187	0.1074
Active current	4766.0	-2.8879	0.0262	-3.3347	-2.9041	-2.8830
Axial force	4766.0	5.3454	0.0306	5.2266	5.3265	5.3516
Motor current phase 1	4766.0	0.0451	0.0174	-0.0040	0.0334	0.0430
Motor current phase 2	4766.0	-0.0266	0.0264	-0.0915	-0.0462	-0.0337
Motor current phase 3	4766.0	0.0140	0.0199	-0.0250	-0.0038	0.0135
Pneumatic pressure	4766.0	832722.3146	1860.0271	827225.1591	831416.5233	832856.7918
Sound pressure	4766.0	-0.0958	0.0342	-0.2819	-0.1183	-0.0971
Velocity	4766.0	-199.9139	0.0503	-200.2197	-199.9290	-199.9053

Loading [MathJax]/jax/output/CommonHTML/fonts/TeX/fontdata.js

In [6]:
```
fig, axes = plt.subplots(3, 4, figsize=(14, 8))
for ax, name in zip(axes.flat, CHANNEL_NAMES):
    ax.hist(df_mean[name], bins=40, color="#4C72B0")
    ax.set_title(name, fontsize=9)
for ax in axes.flat[len(CHANNEL_NAMES):]:
    ax.axis("off")
fig.suptitle("Per-cycle mean value distribution, by channel")
plt.tight_layout()
plt.show()
```



5. Per-channel uncertainty

Key finding: unlike the raw values, the standard uncertainty is **constant across every cycle and every within-cycle reading** for 6 of the 11 channels (acceleration × 3 axes, axial force, pneumatic pressure, sound pressure) — a single calibration-sheet value applied uniformly. The other 5 channels (active current, the 3 motor-current phases, velocity) carry a per-sample uncertainty that varies with the reading itself.

Loading [MathJax]/jax/output/CommonHTML/fonts/TeX/fontdata.js

In [7]:

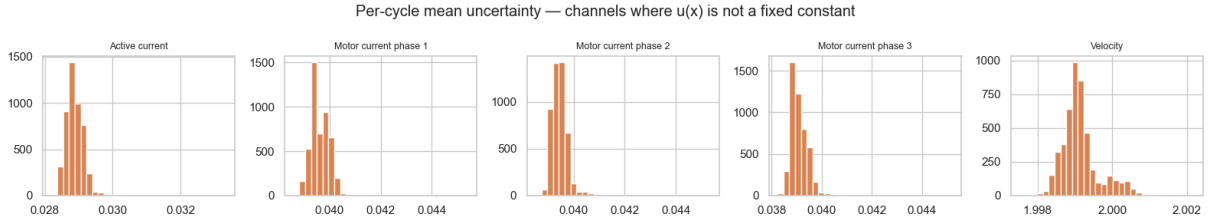
```
unc_summary = []
for i, name in enumerate(CHANNEL_NAMES):
    block = UX[channel_slice(i)]
    val_block = X[channel_slice(i)]
    n_unique = len(np.unique(np.round(block, 10)))
    rel = np.abs(block / np.where(val_block == 0, np.nan, val_block))
    unc_summary.append({
        "channel": name,
        "u constant?": "yes" if n_unique == 1 else "no",
        "u mean": block.mean(),
        "u min": block.min(),
        "u max": block.max(),
        "median |u/x|": np.nanmedian(rel),
    })
unc_df = pd.DataFrame(unc_summary).set_index("channel")
unc_df
```

Out[7]:

channel	u constant?	u mean	u min	u max	median |u/x|
Acceleration plain bearing	yes	2.8319e+00	2.8319e+00	2.8319e+00	3.2803e+00
Acceleration piston rod	yes	2.8319e+00	2.8319e+00	2.8319e+00	7.7770e-01
Acceleration ball bearing	yes	2.8319e+00	2.8319e+00	2.8319e+00	1.0872e+00
Active current	no	2.8879e-02	1.2612e-02	3.9580e-02	1.0000e-02
Axial force	yes	1.2437e-02	1.2437e-02	1.2437e-02	2.3185e-03
Motor current phase 1	no	3.9629e-02	4.0416e-06	8.1029e-02	1.5000e-02
Motor current phase 2	no	3.9433e-02	4.4041e-08	8.0332e-02	1.5000e-02
Motor current phase 3	no	3.9087e-02	2.9153e-07	8.2582e-02	1.5000e-02
Pneumatic pressure	yes	1.6833e+04	1.6833e+04	1.6833e+04	2.0138e-02
Sound pressure	yes	2.7885e-05	2.7885e-05	2.7885e-05	7.2025e-05
Velocity	no	1.9991e+00	1.1909e+00	2.1938e+00	1.0000e-02

Loading [MathJax]/jax/output/CommonHTML/fonts/TeX/fontdata.js

In [8]:
```
variable_channels = [n for n in CHANNEL_NAMES if unc_df.loc[n, "u
constant?"] == "no"]
fig, axes = plt.subplots(1, len(variable_channels), figsize=(16, 3))
for ax, name in zip(axes, variable_channels):
    i = CHANNEL_NAMES.index(name)
    ax.hist(UX[channel_slice(i)].mean(axis=1), bins=30, color="#DD8452")
    ax.set_title(name, fontsize=9)
fig.suptitle("Per-cycle mean uncertainty — channels where u(x) is not a
fixed constant")
plt.tight_layout()
plt.show()
```



6. Degradation trend: channel means vs. consumed lifetime

Since `y` is monotonic in cycle index, plotting each channel's per-cycle mean against `y` shows which sensors trend with wear/degradation and which stay flat. Pearson correlation ranks the channels.

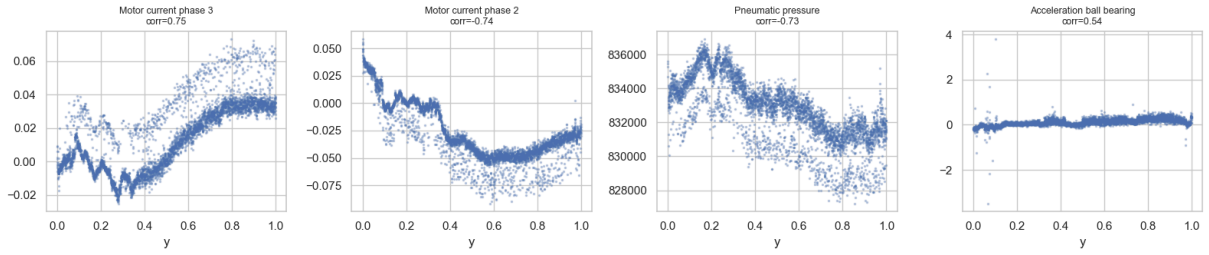
In [9]:
```
corr = df_mean.corrwith(pd.Series(y)).sort_values(key=lambda s: s.abs(),
ascending=False)
corr.to_frame("corr with y")
```

Out[9]:

	corr with y
Motor current phase 3	0.7458
Motor current phase 2	-0.7387
Pneumatic pressure	-0.7297
Acceleration ball bearing	0.5407
Acceleration piston rod	0.5152
Sound pressure	-0.3014
Velocity	-0.2479
Motor current phase 1	-0.1801
Axial force	0.1801
Acceleration plain bearing	0.1200
Active current	0.1019

Loading [MathJax]/jax/output/CommonHTML/fonts/TeX/fontdata.js

In [10]:
```
top = corr.abs().sort_values(ascending=False).index[:4]
fig, axes = plt.subplots(1, 4, figsize=(16, 3.5))
for ax, name in zip(axes, top):
    ax.scatter(y, df_mean[name], s=2, alpha=0.3)
    ax.set_title(f"{name}\ncorr={corr[name]:.2f}", fontsize=9)
    ax.set_xlabel("y")
plt.tight_layout()
plt.show()
```

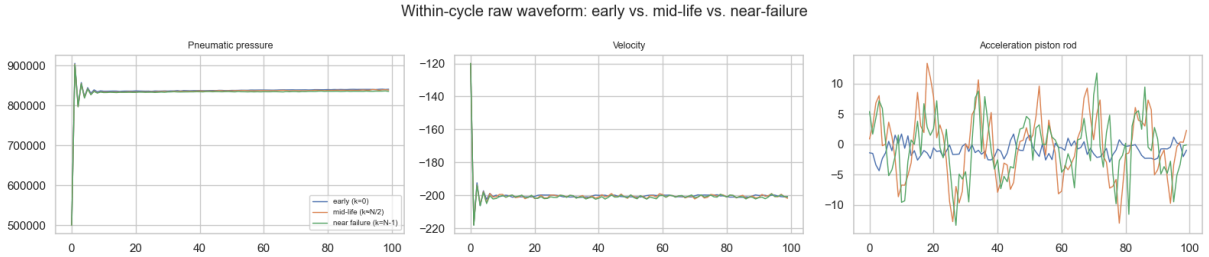


7. Example raw waveforms within a cycle

Beyond per-cycle summaries, each cycle carries 100 raw readings per channel. Comparing an early, mid-life, and near-failure cycle's raw waveform shows shape changes that the per-cycle mean alone can hide.

In [11]:
```
example_cycles = {
    "early (k=0)": 0,
    "mid-life (k≈N/2)": len(y) // 2,
    "near failure (k=N-1)": len(y) - 1,
}
plot_channels = ["Pneumatic pressure", "Velocity", "Acceleration piston rod"]

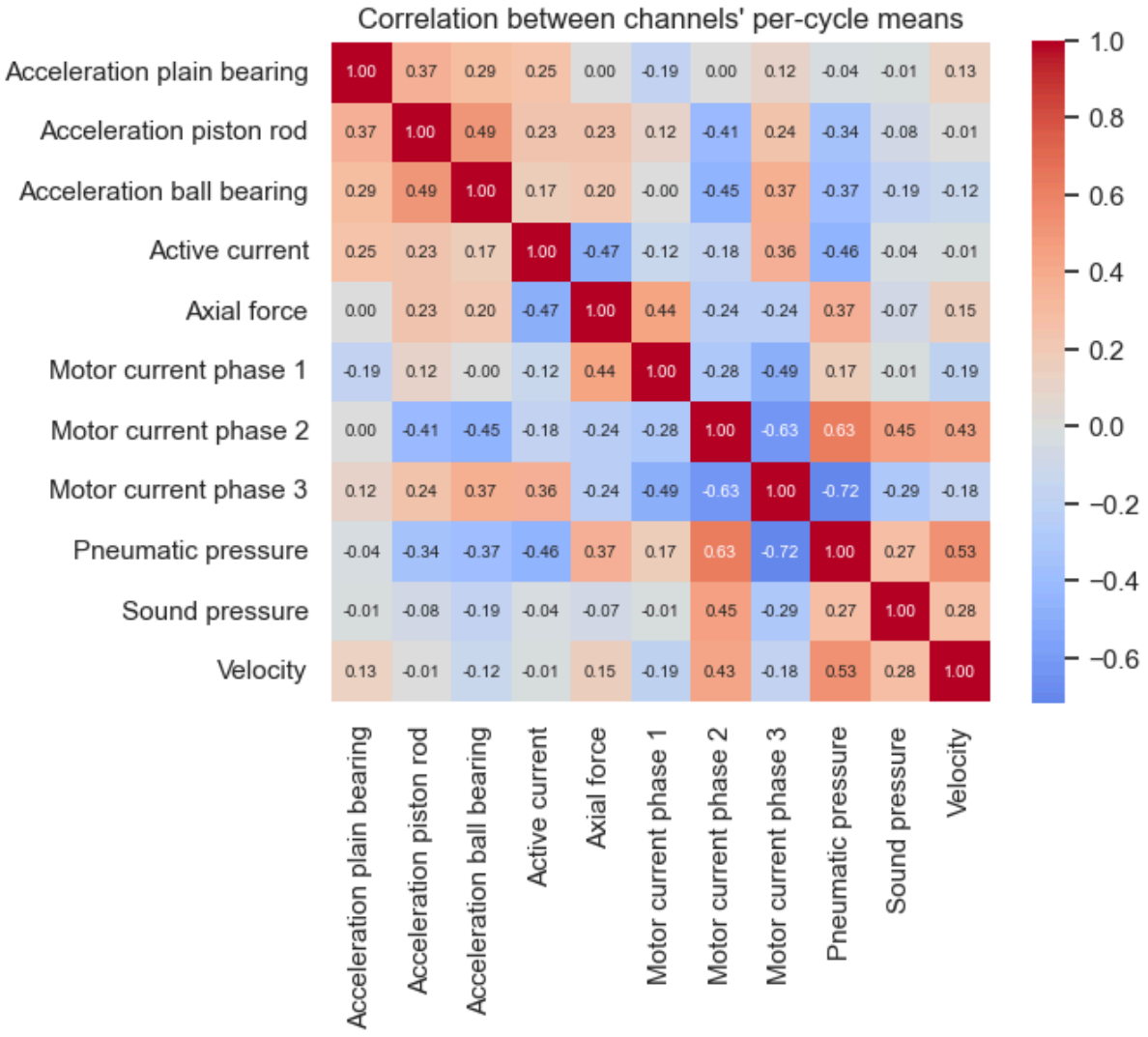
fig, axes = plt.subplots(1, len(plot_channels), figsize=(16, 3.5))
for ax, name in zip(axes, plot_channels):
    i = CHANNEL_NAMES.index(name)
    for label, k in example_cycles.items():
        ax.plot(X[k, i * D:(i + 1) * D], lw=1, label=label)
    ax.set_title(name, fontsize=9)
axes[0].legend(fontsize=7)
fig.suptitle("Within-cycle raw waveform: early vs. mid-life vs. near-failure")
plt.tight_layout()
plt.show()
```



8. Correlation across channels

Loading [MathJax]/jax/output/CommonHTML/fonts/TeX/fontdata.js

In [12]:
```
fig, ax = plt.subplots(figsize=(7, 6))
sns.heatmap(
    df_mean.corr(), annot=True, fmt=".2f", cmap="coolwarm", center=0,
    square=True, ax=ax, annot_kws={"fontsize": 7},
)
ax.set_title("Correlation between channels' per-cycle means")
plt.tight_layout()
plt.show()
```



9. Sanity checks

In [13]:
```
print("NaNs in X:", np.isnan(X).sum(), " | NaNs in UX:",
np.isnan(UX).sum())
print("Infs in X:", np.isinf(X).sum(), " | Infs in UX:",
np.isinf(UX).sum())
zero_var = [name for name in CHANNEL_NAMES if df_mean[name].std() == 0]
print("Channels with zero variance across cycles:", zero_var or "none")
```

```
NaNs in X: 0  | NaNs in UX: 0
Infs in X: 0  | Infs in UX: 0
Channels with zero variance across cycles: none
```

Loading [MathJax]/jax/output/CommonHTML/fonts/TeX/fontdata.js

10. Appendix — model output (`case_study_A_results.csv`)

Quick look at the pipeline's output artifact for continuity with the raw-data analysis above. This is *not* a replacement for `make_validation_figure.py` (the paper's actual validation figure) — just a sanity glance at the shape of the results.

In [14]:
```
results = pd.read_csv("case_study_A_results.csv")
results.columns = results.columns.str.lstrip("# ")
print(results.shape)
results.describe().T
```

```
(953, 7)
```

Out[14]:

	count	mean	std	min	25%	50%	75%	max
y_true	953.0	0.5031	0.2911	0.0000	0.2558	0.4823	0.7650	0.9994
y_hat	953.0	0.4968	0.2932	-0.1303	0.2436	0.4722	0.7582	1.1274
u_physical_lpu	953.0	0.6413	0.1578	0.3355	0.5106	0.6116	0.7844	1.0779
u_physical_mc	953.0	0.9841	0.0826	0.7916	0.9216	0.9747	1.0575	1.1944
u_training	953.0	0.0493	0.0260	0.0157	0.0322	0.0416	0.0574	0.2138
u_c	953.0	0.6437	0.1579	0.3369	0.5132	0.6133	0.7856	1.0822
covered_k2	953.0	1.0000	0.0000	1.0000	1.0000	1.0000	1.0000	1.0000

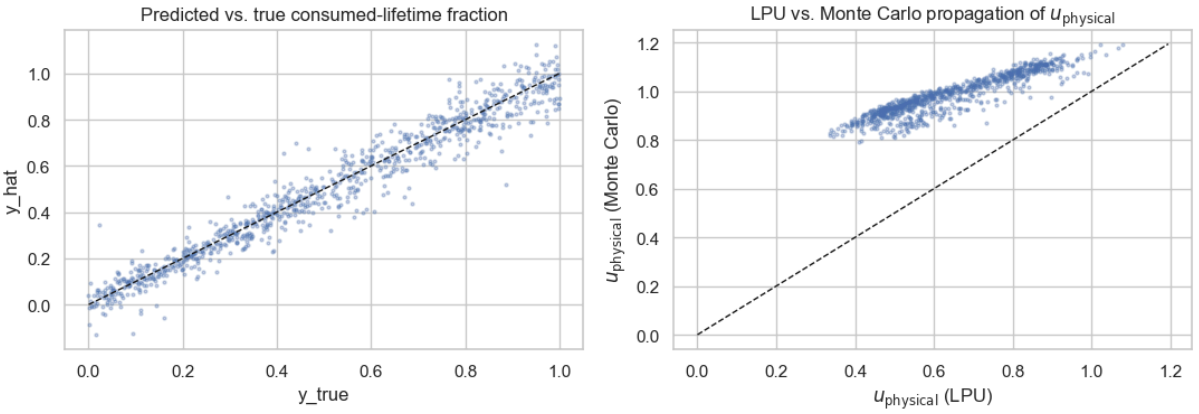
Loading [MathJax]/jax/output/CommonHTML/fonts/TeX/fontdata.js

In [15]:
```
fig, axes = plt.subplots(1, 2, figsize=(11, 4))

axes[0].scatter(results["y_true"], results["y_hat"], s=4, alpha=0.3)
lims = [0, 1]
axes[0].plot(lims, lims, "k--", lw=1)
axes[0].set_xlabel("y_true"); axes[0].set_ylabel("y_hat")
axes[0].set_title("Predicted vs. true consumed-lifetime fraction")

axes[1].scatter(results["u_physical_lpu"], results["u_physical_mc"], s=4,
alpha=0.3)
lim = max(results["u_physical_lpu"].max(), results["u_physical_mc"].max())
axes[1].plot([0, lim], [0, lim], "k--", lw=1)
axes[1].set_xlabel(r"$u_{\mathrm{physical}}$ (LPU)");
axes[1].set_ylabel(r"$u_{\mathrm{physical}}$ (Monte Carlo)")
axes[1].set_title(r"LPU vs. Monte Carlo propagation of
$u_{\mathrm{physical}}$")

plt.tight_layout()
plt.show()
```



Loading [MathJax]/jax/output/CommonHTML/fonts/TeX/fontdata.js

Summary

- **Shape**: 4,766 run-to-failure cycles × 11 sensor channels × 100 readings/cycle; no missing or infinite values; no zero-variance channels.
- **Target** `y` is a deterministic ramp (cycle index / N) — zero label uncertainty by construction.
- **Uncertainty structure**: 6 of 11 channels (acceleration × 3 axes, axial force, pneumatic pressure, sound pressure) carry a single, constant calibration uncertainty applied to every cycle and reading; the other 5 (active current, the 3 motor-current phases, velocity) carry a per-sample uncertainty that varies with the reading itself.
- **Degradation trend**: motor current (phases 2 and 3) and pneumatic pressure correlate most strongly with consumed-lifetime fraction (|corr| ≈ 0.73–0.75), followed by the ball-bearing and piston-rod accelerometers (≈ 0.52–0.54); active current and the plain-bearing accelerometer are nearly uncorrelated with wear. Within-cycle waveform *shape* (not just the mean) also visibly changes from early to near-failure cycles for pressure, velocity and acceleration.
- These are descriptive, raw-data statistics only. The trained-model *sensitivity* ranking reported in `case_study_A_report.md` §7/§9 (Pressure 51.8%, Velocity 25.2%, Acceleration-x 11.5% of first-layer sensitivity) combines normalized uncertainty with the learned network weights, and is not simply reproduced from raw channel variance or correlation with `y`.

Loading [MathJax]/jax/output/CommonHTML/fonts/TeX/fontdata.js